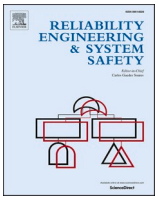
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Online surface temperature prediction and abnormal diagnosis of lithium-ion batteries based on hybrid neural network and fault threshold optimization

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ABSTRACT

Online diagnosis of abnormal temperature is vital to ensure the reliability and operation safety of lithium-ion batteries, and this study develops a hybrid neural network and fault threshold optimization algorithm for their online surface temperature prediction and abnormal diagnosis. To be specific, a hybrid neural network incorporating convolutional neural network and long short-term memory neural network is firstly employed to predict the battery temperature, and a residual monitor is designed to track the deviation between the measure and the prediction. Then, the acquired residual is compared with the fault threshold to diagnose whether the battery temperature is abnormal. Moreover, to improve the correctness and reliability of fault diagnosis, a fault threshold optimization algorithm based on the receiver operating characteristic curve is defined to automatically find the optimal fault threshold. The accuracy and reliability of temperature prediction is verified under various aged state and temperatures, as well as different battery types. The abnormal experiment validation on three datasets reveals that the proposed method can diagnose and unwind temperature fault warning timely and reliably. Additionally, the average execution time of each prediction and diagnosis is less than 3.5 ms, manifesting the real-time application capability of the proposed method.

1. Introduction

To alleviate fuel crisis and reduce environmental pollution, lithium-ion batteries, as a promising energy storage media, are widely furnished in electric vehicles (EVs) and power grids due to their high energy density, long life and low cost [1]. As well known, the reliability and safety management of lithium-ion batteries are the basis for their efficient and effective engineering applications [2]. However, many electrical and thermal faults occur inside the battery, discounting the operation safety and reliability [3]. Generally, the most critical battery malfunctions include over-heating, over-charging, over-discharging, internal and external short circuits, accelerated degradation, swelling, electrolyte leakage, and thermal runaway. These failures showcase diverse modes, sophisticated mechanisms and intertwined relationships, and there does not exist a one-size-fits-all solution for all the failures in battery systems [4]. How to diagnose the battery fault reliably and timely is of great significance for not only battery's normal operation, but also the vehicle's safety.

A variety of diagnosis algorithms have been widely investigated to achieve timely fault diagnosis occurring in lithium-ion batteries. To the authors' knowledge, these diagnosis methods can be roughly categorized into knowledge-based, model-based and data driven methods [5]. Knowledge-based methods generally exploit and explore prior expert knowledge to design fault diagnosis rules based on external measures. Although this kind of approaches do not entail modeling, the priori knowledge acquisition and rules formation are quite time- and labor-consuming [6]. Model based methods, as the name implies, employ mathematical models to mimic the battery internal electrochemical reactions, and leverage various identification algorithms to forecast the parameters that can characterize the battery fault. The predicted fault parameters are usually compared with measures, and then the residual signals are obtained and analyzed for fault diagnosis.

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Ref. [7] establishes a first-order equivalent circuit model (ECM) with the voltage as input and the current as output, and the least squares method combined with the unscented Kalman filter (UKF) is leveraged to detect current sensor faults. In [8], a thermal-electrical model is established for the lithium-ion battery pack, and particle filter (PF) is applied to predict the temperature and voltage of the battery pack. On this basis, the faults of voltage, current, temperature sensors are detected by the residual assessment. However, these model-based approaches depend excessively on the accuracy of the established model, and require sufficient priori-knowledge of electro-chemical reactions mechanisms and much modeling effort [9].

Currently, numerous studies have shown that data-driven methods without modeling are not only extensively employed in battery state of health (SOH) prognosis [10] and capacity estimation [11], but also well implemented in battery fault diagnosis [12]. Ref. [13] develops the least-square support vector regression (LS-SVR) to forecast the battery voltage variation and detect the potential inconsistency fault of battery pack. In [14], a wavelet neural network (NN) model is constructed to detect the connection fault of battery pack. With the rapid development of artificial intelligence, deep learning algorithms, such as deep NN (DNN), convolutional NN (CNN) and recurrent NN (RNN), show more powerful application potentials in the field of feature extraction and regression prediction [15]. In [16], a DNN based on stacked sparse autoencoder is employed to diagnose fault for the solid oxide fuel cell system. Ref. [17] resorts to signal imaging and CNN to supply judgements on fault type and grade of battery packs. In [18], long short-term memory (LSTM) NN is exploited to better capture the fault related time series information, thereby estimating the current distribution and diagnose the inconsistency fault for parallel-connected lithium-ion batteries.

Generally, in some failure scenarios such as short circuit faults, inconsistency faults and connection faults, the battery will generate abnormal heat, and correspondingly the temperature will rise [4]. Furthermore, when the raised temperature sparks the abnormal heat generation and exothermic reactions, thermal runaway, and even fire and explosion, may be triggered [19]. Obviously, excessive battery temperature increase is a direct representation for thermal runway [20], and online prediction and diagnosis of battery temperature can contribute to safety and reliable operation of batteries. In [21], the electrochemical impedance spectrum is employed for online temperature monitoring. Ref. [22] develops an electrical-thermal model to achieve the real-time temperature prediction. In [23], online temperature estimation is achieved by combining extended Kalman filter (EKF) and a NN model. To diagnose the battery temperature fault, Ref. [24] constructs an electrothermal model and leverages LSTM NN to forecast the battery surface temperature in real time, achieving early warning of temperature. Based on the literature review, the limitations of existing studies can be summarized as follows. First, most of the reported methods for temperature prediction accounts for limited ranges of operating conditions, e.g., fixed operating temperature, fixed aging degree or single battery type. However, in practice, lithium-ion battery not only operates over a wide temperature range, but also encounters random charging and discharging conditions and undergoes aging. In these situations, the battery temperature variation is more complex. Therefore, it is imperative to design a robust temperature prediction method that can adapt to complex operation situations. Second, online abnormal temperature fault, to the authors' knowledge, needs to be further investigated. Third, the referred methods lack theoretical basis for threshold selection, and the maximum error of model is simply selected as the threshold, which may deteriorate the diagnosis precision or late judgement.

To remedy the above imperfections, a temperature prediction and abnormal diagnosis algorithm based on hybrid NN and fault threshold optimization is developed. First, a CNN-LSTM network is developed to forecast battery temperature online by fully exploiting its advantages of extracting spatial and temporal features. Then, to determine the appropriate input parameters for the CNN-LSTM network, an equivalent circuit thermal model is constructed, and the influence of each parameter on temperature variation is addressed. Next, a fault threshold optimization method is presented to optimize the fault threshold in the diagnosis strategy based on the receiver operating characteristic (ROC) curve, which is applied to compare the deviation between the measurement and the prediction from the CNN-LSTM network and diagnose whether the temperature is normal. The validation results under complex operation scenarios showcase that the CNN-LSTM network can achieve satisfactory temperature prediction accuracy, and the proposed diagnosis strategy can timely and reliably diagnose the temperature anomaly. This study is committed to making contributions in the following three aspects. (1) A CNN-LSTM network is developed to efficiently excavate time-series and spatial features in battery operation data. The accuracy and reliability of CNN-LSTM network for battery temperature prediction is comprehensively evaluated with two different types of batteries under complex operation scenarios including different operating temperatures and different aging states. (2) A residual monitor is developed to track the residual of prediction and measurement surface temperature, thereby judging whether it exceeds the threshold and triggers a temperature fault. (3) A fault threshold optimization algorithm based on the ROC curve is presented for optimal threshold determination and fault diagnosis for abnormal temperature.

The rest of this study is arranged as follows. Section II introduces the time-series characteristics of battery temperature, and presents the online temperature prediction framework and the abnormal diagnosis strategy. Section 3 details the data collection process and experimental studies. Next, the performances of temperature prediction and fault diagnosis is evaluated in Section IV. Finally, the main conclusions are drawn in Section V.

2. Online temperature prediction and diagnosis

2.1. Temperature variation analysis

To analyze the battery temperature variation process, an equivalent circuit thermal model is exploited in this study, as shown in Fig. 1. It is assumed that the temperature within battery cell is uniform. On this account, the equivalent circuit thermal model for the heat transfer process can be formulated [25], as:

$$\begin{cases} C_c \dfrac{dT_i}{dt} = Q - \dfrac{T_i - T_s}{R_i} \\ C_s \dfrac{dT_s}{dt} = \dfrac{T_i - T_s}{R_i} - \dfrac{T_s - T_a}{R_e} \end{cases} \tag{1}$$

where C_c and C_s respectively denote the heat capacity of the core and surface. T_i and T_s are the internal and surface temperature. T_a represents the ambient temperature. R_i and R_e mean the internal and external thermal resistances. Q represents the heat generation, and can be

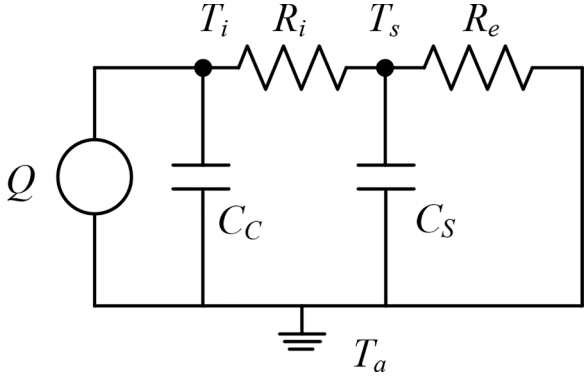


Fig. 1. Equivalent circuit thermal model.

deduced, as:

$$Q = I(U_T - U_{OCV}) \tag{2}$$

where I is the battery current, U_T and U_{OCV} denote the battery terminal voltage and open circuit voltage (OCV). Combining (2) and (1), the state function of the battery temperature variation can be presented, as:

$$\begin{bmatrix} \dot{T}_i \\ \dot{T}_s \end{bmatrix} = \begin{bmatrix} -\frac{1}{C_c R_i} & \frac{1}{C_c R_i} \\ \frac{1}{C_c R_i} & -\frac{2}{C_c R_i} \end{bmatrix} \begin{bmatrix} T_i \\ T_s \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{C_c} \\ \frac{1}{C_s R_e} & 0 \end{bmatrix} \begin{bmatrix} T_a \\ I(U_T - U_{OCV}) \end{bmatrix} \tag{3}$$

Although the above thermal model can simply describe the temperature variation, the model parameters need to be identified in advance to achieve accurate temperature prediction. From the above investigation, it is clearly found that the battery temperature prediction is typically a time series problem. To tackle it, the CNN-LSTM network is developed to estimate the battery temperature. To find the appropriate input parameters for the CNN-LSTM network, the relationship between each model parameter and the surface temperature is analyzed. As can be observed from (3), the battery temperature is related to its inherent chemical parameters (including R_i, R_e, C_s and C_c), terminal voltage, OCV, operation current and previous moment temperature. Generally, R_i, R_e, C_s and C_c feature highly nonlinear relationships with the terminal voltage, operating current and state of charge (SOC), and thus these parameters can be leveraged to characterize the variation of internal parameters. Furthermore, online measure of OCV remains challenging. Fortunately, there exists a direct mapping relationship between SOC and OCV, and thus SOC can be hired to represent OCV for online heat analysis. Additionally, as described in [26], the ohmic resistance in traditional ECM increases with SOC decrease. As well known, the increase of internal resistance will further lead to increase in battery temperature. Obviously, the battery SOC poses a significant effect on battery temperature. To sum up, the battery SOC, current, voltage at the current moment and previous moment temperature are selected as the inputs to the CNN-LSTM network to predict the battery temperature at the current moment.

2.2. Online temperature prediction

Currently, LSTM and CNN are widely hired to handle temporal and spatial mapping problems, respectively. In this study, these two algorithms are integrated to elicit temporal and spatial features of battery historical working data for reliable temperature prognosis. As shown in Fig. 2, the established CNN-LSTM network consists of five layers, namely the input layer, convolution layer, max pooling layer, LSTM layer and the fully connected layer. The inputs of the network include current, voltage, and SOC at the current moment, and previous moment temperature, and the output is the temperature at the current moment. Moreover, the CNN consists of convolution and max pooling layer and shows strong abilities in capturing the spatial correlation among the inputs [27]. As presented in [28], spatial features refer to the characteristics obtained by the convolution kernels sliding on the inputs in CNN. From Fig. 2, it can be found that the inputs of each time step are one-dimensional, and the convolution kernel only slides over the four input parameters and performs convolution operations. Therefore, the spatial feature of this study can be described as the interactions and relationships among the four input parameters. By assuming that the input matrix is In_t, the spatial features obtained after the convolution operation is expressed, as:

$$O_t = f(In_t \otimes \alpha + \beta) \tag{4}$$

where α and β mean the weight and bias of the convolution layer, $f(\cdot)$ denotes the activation function, $\otimes$ stands for the convolution operation. Next, the maximum pooling layer is applied to extract the most critical spatial features in O_t to avoid over-fitting caused by excessive dimension, as:

$$p_t = max\left\{O_t^{(n\times 1)} m(n,1)\right\} \tag{5}$$

where $m(n,1)$ denotes the window size in the pooling process, and p_t represents the matrix after the pooling operation. Then, p_t is treated as the input of LSTM layer, which can extract temporal features including the long-term and short-term dependencies by input gate (i), output gate (o), forget gate (f), memory unit (c) and hidden state (h). For an input vector p_t, the output of the LSTM unit is formulated, as:

$$\begin{cases} i_t = \sigma(W_i p_t + W_i h_{t-1} + b_i) \\ f_t = \sigma(W_f p_t + W_f h_{t-1} + b_f) \\ c_t = f_t * c_{t-1} + i_t * tanh(W_c p_t + W_c h_{t-1} + b_c) \\ o_t = \sigma(W_o p_t + W_o h_{t-1} + b_o) \\ h_t = o_t * tanh(c_t) \end{cases} \tag{6}$$

where σ means sigmoid function, W and b denote the weight and bias of the corresponding gate or unit, and $*$ represents the dot product operation. In addition, a fully connected layer is deployed behind the LSTM layer to collect the feature, and each neuron in this layer is connected to all the activations in the previous layer.

As can be observed, when the inputs sequentially pass through the CNN layer and the LSTM layer, the spatial and temporal features in the four input parameters can be efficiently learnt and extracted. In this study, the hyperparameters of the CNN-LSTM network are determined by trial-and-error, and finally the number of convolutional kernels, the convolutional kernel size, the polling size, the number of LSTM neurons and the max number of epochs after optimization are set to 3, 1, 1, 200 and 200, respectively. During the training process, the Adam algorithm is employed to optimize the model parameters. It is a stochastic gradient optimizer, which not only optimizes the gradient of the stochastic objective function, but also improves the computational efficiency significantly while entailing less storage [29]. It is worth noting that the proposed CNN-LSTM network training process is obtained offline, but when the model training is completed, the current moment temperature can be predicted online once the present moment current, voltage, SOC and previous moment temperature of the battery are acquired. Additionally, combining the temperature prediction values with the

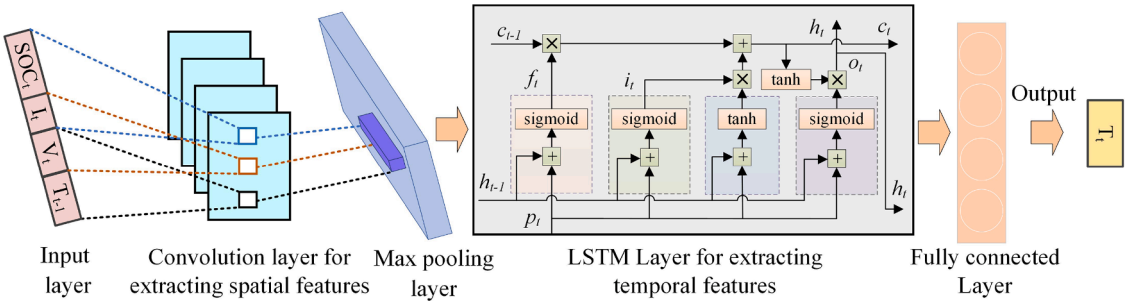


Fig. 2. Structure of the developed CNN-LSTM network.

diagnosis strategy can realize online abnormal temperature diagnosis, which will be described in detail in the next part.

2.3. Online abnormal temperature diagnosis strategy

In this part, the diagnosis strategy utilized for online recognizing abnormal temperature is elucidated. When batteries work in reasonable state, their temperature is kept within a certain range. On this premise, abnormal variations in battery temperature can be detected by comparing the measures with the predictions. The specific implementation process for online discerning abnormal temperature is sketched in Fig. 3. First, the proper input parameters of the CNN-LSTM network are determined by the equivalent circuit thermal model. Then, the CNN-LSTM network is trained offline using the experimental data, and the trained model is implemented to forecast the temperature online under various complicated operation conditions. Next, a residual monitor is developed to calculate the residual value between the measurement and the prediction. Subsequently, the ROC curve is employed to automatically select the optimal battery temperature fault threshold for improving the correct fault diagnosis rate. After that, the battery abnormal temperature is diagnosed online by combining the obtained temperature residuals and the optimal fault threshold. Specifically, if the residual value exceeds the threshold value, the battery temperature is identified abnormal.

From the above process, one can find that the selection of threshold is critical for fault judgment. If the threshold is too loose, the fault will be misinformed, or else if the threshold is too strict, the fault cannot be detected in time. In order to achieve in time and reliable abnormal temperature diagnosis, a fault threshold optimization algorithm is proposed via the ROC curve in this study. The ROC curve shows the true positive rate (TPR) of a classifier against its false positive rate (FPR), where the TPR is the percentage of subjects correctly classified as positive over the total number of positives, as:

$$TPR = \frac{TP}{TP + FN} \tag{7}$$

where TP denotes the number of true positives, and FN means the number of false negatives. While the FPR means the percentage of the subjects correctly classified as true over the total number of negatives, as:

$$FPR = \frac{FP}{FP + TN} \tag{8}$$

where FP represents the number of false positives, and TN stands the number of true negatives. It can be found these two values range from zero to one and are relevant to the classification threshold, which determines whether the object is correct. Therefore, the TPR and FPR of each threshold can be examined by setting different thresholds, and the optimal threshold can be obtained by screening the maximum TPR and minimum FPR. To compare the performance of each classifier, a scalar is considered to evaluate the ROC performance. A common method is to calculate the area under the ROC curve, referred to as the area under the curve (AUC). Since AUC is a part of the unit square area, its value is limited within 0 and 1. However, since the diagonal line between (0, 0) and (1, 1) is generated by random guess and its area is 0.5, the AUC of any real classifier should not be less than 0.5 [30]. Furthermore, it is known that the closer the ROC curve to the upper left corner, the closer the area of the AUC to 1, showcasing that the classifier is more accurate in determining positive and negative samples.

3. Experimental study

In this study, two different source data (labeled as dataset 1 and dataset 2) are collected to appraise the proposed temperature prediction

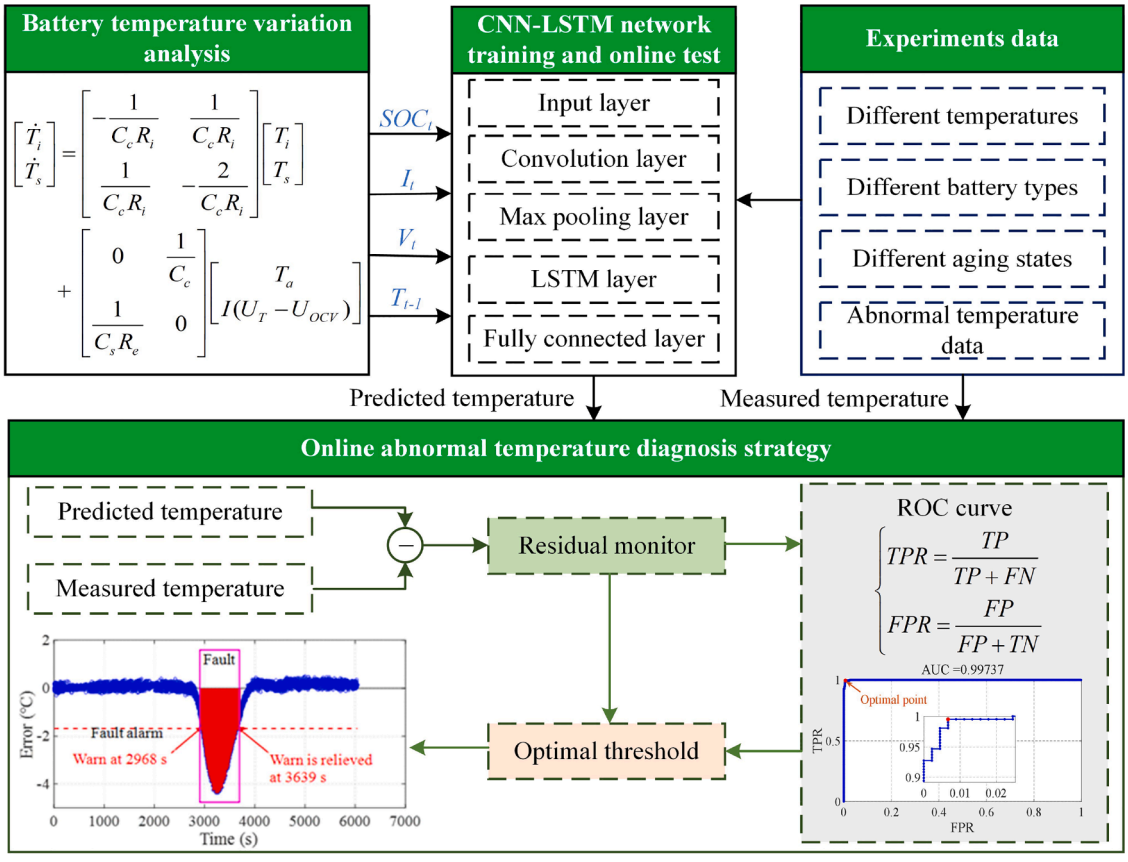


Fig. 3. Framework of the proposed abnormal temperature diagnosis method.

and abnormal temperature diagnosis method. Table 1 shows the specifications of the experimental batteries. The battery materials for the two datasets are different, one is made up of $LiFePO_4$ and the other one is $LiNiCoAlO_2$. From this point of view, the generality and reliability of the proposed temperature prediction method can be better validated. The dataset 1 is tested in the laboratory and consists of three LiFePO4 batteries (called cells 1 to 3), all of which are tested in a thermal controlled chamber with a cycler.

To investigate the environmental suitability and reliability of the proposed method, cells 1 to 3 are respectively experimented at operation temperatures of 40 °C, 10 °C and 30 °C, and a series of hybrid drive cycle tests, namely, urban dynamometer driving schedule (UDDS) and federal urban driving schedule (FUDS) cycles, are repeatedly performed. In these tests, the battery is fully charged with 0.5C current (where C denotes the nominal capacity) constant current-constant voltage (CC-CV) charging protocol, and the specific dynamic profile is implemented until the terminal voltage decreases to 2.0 V. Moreover, a 1C discharge is implemented to calibrate the battery capacity after every 30 cycles. Obviously, we can collect data on the different stages of cell aging as the number of cycles increases. To be specific, cells 1 to 3 undergo 598 cycles, 1770 cycles and 1210 cycles, and their capacity decreases from 4Ah to 3.03Ah, 3.56Ah and 1.65Ah, respectively. Note that a thermocouple is attached to the surface of the battery, and each dataset is sampled with the frequency of 1 Hz. One example of the charging and discharging profiles of cell 3 is depicted in Fig. 4 (a). As can be found, the battery experiences four stages, i.e., discharging, charging, resting and discharging. Moreover, although we set the operating temperature to 30 °C, the battery surface temperature is not kept unchanged, but varies rapidly with the operations and is much greater than 30 °C. This is because the battery is shelved for only 10 min in each cycle, leaving insufficient time to recover to operation temperature during successive cycle experiments. Additionally, when the battery aging gradually deepens, the battery internal resistance increases [31], and also causes additional battery temperature rise. The dataset 2 is collected from [32], which include one $LiNiCoAlO_2$ battery (labeled cell 4). Similarly, the battery is fully charged according to the CC-CV scheme, then discharged by a series of driving cycles, including UDDS, Los Angeles 92 cycle (LA92), supplemental federal test procedure driving schedule (US06), and highway fuel economy driving schedule (HWFET). Fig. 4 (b) shows the profiles of battery current, voltage and temperature for cell 4. It is clearly noticed that the operating temperature of this cell during the test changes over time, and ranges from a low temperature -20 °C to 20 °C.

4. Results and discussions

In this section, the accuracy and reliability of the proposed method is firstly validated at different operation temperatures, and the prediction results are compared with different algorithms. Next, the versatility of the proposed method is evaluated based on different types of batteries. Then, the temperature applicability of the proposed method for aged cell is investigated. Finally, the battery abnormal temperature is diagnosed. Notably, the ratio of the model training set and test set is 7:3. Furthermore, the temperature prediction accuracy of the proposed method can be expressed by the root mean square error (RMSE), max absolute error (MAX) and mean absolute error (MAE), as:

Table 1
Specifications of the experimental batteries.

Source	Dataset 1 (cells 1 to 3)	Dataset 2 (cell 4)
Type	21700	18650
Material	$LiFePO_4$	$LiNiCoAlO_2$
Nominal voltage (V)	3.25	3.6
Nominal capacity (Ah)	4	2.9
Operation voltage (V)	2.0-3.65	2.5-4.2

$$\begin{cases} RMSE = \sqrt{\frac{1}{L}\sum_{t=1}^{L}\left(T_t - T_t^*\right)^2} \\ MAX = max\left(|T_t - T_t^*|\right) \\ MAE = \frac{1}{L}\sum_{t=1}^{L}|T_t - T_t^*| \end{cases} \tag{9}$$

where L means the sample length of the test data, T_t^* and T_t indicate the reference and predicted value of temperature at time t.

4.1. Temperature prediction at different operation temperatures

Since the environment shows an obvious effect on temperature prediction, the proposed CNN-LSTM network is firstly validated using dataset 1 obtained under different ambient temperature scenarios. Note that this part utilizes the initial aging datasets of cells 1 to 3. First, the proposed method is validated using an initial aging dataset from cell 1 at 40 °C, and the prediction results are presented in Fig. 5 (a) and (b). As can be clearly observed, even though the battery is experimented at 40 °C, the surface temperature exhibits a fluctuated and increasing trend thanks to battery charging and discharging. This phenomenon provides uncertainty and unpredictability in temperature prediction. Fortunately, from the prediction results, the battery temperature prediction curve of the proposed method can effectively track the measured temperature curve, and the errors are within ±0.3 °C. Then, the predicted results using the initial aging data of cell 2 at 10 °C are shown in Fig. 5 (c) and (d). It can be observed that although the surface temperature of cell 2 varies between 10 °C and 20.2 °C, the prediction curve of the proposed method is quite close to the measured temperature curve. As listed in Table 2, the RMSE, MAX and MAE for cell 2 are respectively 0.11 °C, 0.58 °C and 0.085 °C, manifesting the prediction effectiveness of the proposed algorithm.

Next, an initial aging dataset from cell 3 at 30 °C operating environment is adopted to verify the proposed method. To further evaluate the performance of the proposed method, six commonly machine learning algorithms are hired for comparison, including relevance vector machine (RVM), support vector machine (SVM), random forest (RF), BPNN, CNN and LSTM NN. The detailed comparisons result of different algorithms is demonstrated in Fig. 5 (e) and (f). From Fig. 5 (e), the temperature prediction curves of all the comparison methods show significant fluctuations, indicating that these methods are intractable to track the temperature variation trajectory accurately. Moreover, according to Fig. 5 (f), we can find that the error distribution raised by the proposed method is more concentrated and closer to the zero, while the error by other methods is more dispersed. This is because that traditional machine learning methods, including RVM, SVM, RF and BPNN, only take the present measure signals as the inputs, and their predictions cannot adaptively accommodate the changes of historical information. Furthermore, CNN and LSTM NN cannot capture both spatial and temporal features in the inputs to better estimate the battery temperature. By contrast, the proposed network incorporates the advantages of CNN and LSTM NN to efficiently excavate spatial and temporal features from the historical information and map these features to the temperature trajectory.

To compare the estimation results more intuitively, Table 2 tabulates the RMSE, MAX and MAE by all the methods. It can be observed that the temperature prediction performance of the proposed method outperforms that of all the comparison methods, and the BPNN shows the worst performance among all the candidate methods. Specifically, the RMSE of RVM, SVM, RF, BPNN, CNN and LSTM NN is 0.14 °C, 0.17 °C, 0.24 °C, 0.38 °C, 0.23 °C and 0.15 °C, respectively, while the RMSE of the proposed method is only 0.10 °C under the same situation. Meanwhile, the MAX and MAE of the proposed method are 0.57 °C and 0.082 °C, which are significantly lower than the six comparison

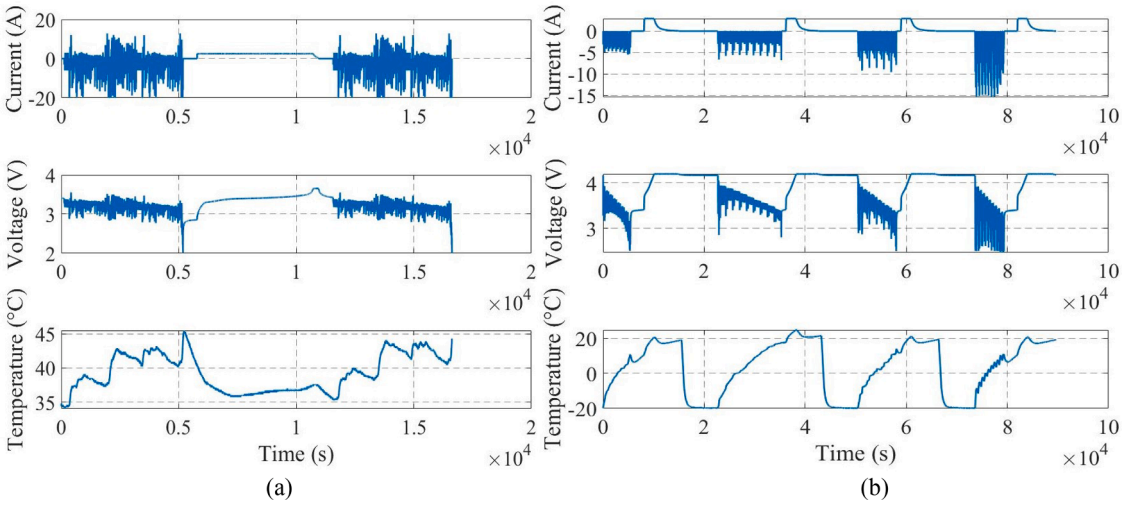


Fig. 4. Driving cycle profiles. (a) Driving cycle profiles of cell 3. (b) Driving cycle profiles of cell 4.

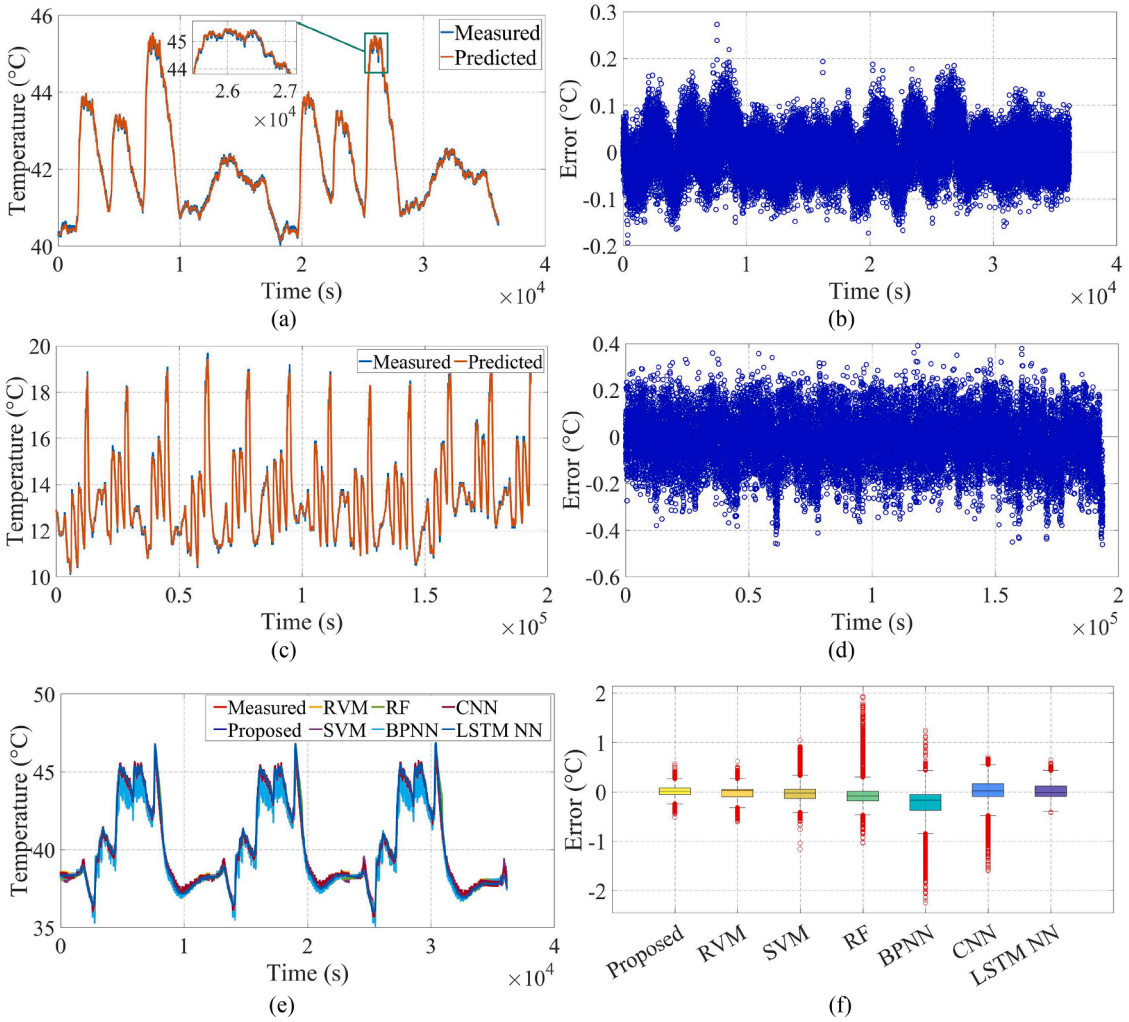


Fig. 5. Comparison results with different methods. (a) Temperature profiles of cell 1. (b) Temperature prediction errors of cell 1. (c) Temperature profiles of cell 2. (d) Temperature prediction errors of cell 2. (e) Temperature profiles of cell 3. (f) Temperature prediction errors of cell 3.

methods. Furthermore, due to the same battery material and similar working state, the temperature prediction results reported in the literature are compared with that by the proposed method, and the statistic errors are also shown in Table 2. In the referenced literature, the GRU NN, the model-based EKF and UKF, and the impedance measurement method are employed to estimate battery temperature. It can be found that the minimum RMSE from the literature is 0.17 °C, which is obviously larger than that of the proposed method. In summary, the

Table 2
Comparison of different temperature prediction methods.

Method	Cycle	Temperature	Battery type	RMSE	MAX	MAE
Proposed	UDDS+	40 °C (cell 1)	LiFePO4	0.047	0.27	0.038
	FUDS	10 °C (cell 2)		0.11	0.58	0.085
		30 °C (cell 3)		0.10	0.57	0.082
RVM				0.14	0.61	0.10
SVM				0.17	1.16	0.12
RF				0.24	1.93	0.16
BPNN				0.38	2.25	0.27
CNN				0.23	1.59	0.17
LSTM NN				0.15	0.65	0.12
GRU NN [25]	FUDS	25 °C	LiFePO4	0.17	/	/
		50 °C		0.18	/	/
EIS + Doubel EKF [33]	Artemis cycle	8 °C	LiFePO4	1.34	/	/
Electrochemical-thermal-NN model + UKF [34]	FUDS	40 °C	LiFePO4	0.24	/	/
Online impedance measurements [35]	10 kHz – 1 Hz	10 °C–55 °C	LCO Li-polymer	1.41	/	/

comparison results show the satisfactory accuracy of the proposed method and prove its superiority and reliability in temperature prediction.

4.2. Temperature prediction on different battery types

The versatility of the temperature prediction method to different battery types is examined on initial aging dataset from cell 4. Note that the experimental target of this dataset is 18650 LiNiCoAlO2 battery, which is different from that of cells 1 to 3, and it operates under continuous time-varying temperature scenarios from a low temperature -20 °C to 20 °C. From this point of view, this case not only demonstrates the performance of the CNN-LSTM model across battery types, but also shows its adaptivity in a varying temperature environment, especially under low temperature driving conditions. The prediction results for cell 4 is depicted in Fig. 6. It can be noticed that the predicted temperature values closely follow the measured values with minimal fluctuations as the battery temperature changes from -20 °C to 14.3 °C. Moreover, the prediction error roughly tends to decrease and then increase with the operating temperature, and the maximum value occurs in low temperature environments. Additionally, we can find that the RMSE, MAX and MAE of cell 4 are all slightly larger than those of cells 1 to 3. The reasons lie in that, firstly, the temperature prediction for low temperature conditions remains challenging [36], and on the other hand, the data has a wider range of temperature variations, leading to more challenges in accurately predicting temperatures. Overall, although the temperature prediction of cell 4 is not as satisfactory as that of cells 1 to 3, the results are sufficient to demonstrate that the proposed method can achieve effective and reliable temperature prediction across different cell types and over a wide range of time-varying temperature.

4.3. Temperature prediction on aged cell

In this part, the capability of the proposed temperature prediction method is investigated on the aging datasets of cells 1 to 3. As described in Section III, the capacity of cells 1 to 3 respectively declines from 4Ah to 3.03Ah, 3.56Ah and 1.65Ah after repeated cycle experiments. It is clearly visible that three batteries deteriorate differently, and cell 3 deteriorates to a much greater extent than cells 1 and 2. The prediction results are demonstrated in Fig. 7. To be specific, Fig. 7 (a) and (b) reveal that the proposed method can effectively track the temperature change, even though the surface temperature of cell 1 varies within the range of 43.4 °C–45.8 °C. And its three-evaluation metrics, including the RMSE, MAX and MAE, are 0.03 °C, 0.16 °C and 0.03 °C, respectively. From Fig. 7 (c), it can be noticed that the surface temperature of cell 2 varies from 16.7 °C to 25 °C with an increase of 8.3 °C. Clearly, the temperature variation range of cell 2 is significantly larger than that of cell 1, and consequently its temperature prediction remains more difficult than that of cell 1. From Fig. 7 (d), the RMSE, MAX and MAE of cell 2 are as low as 0.10 °C, 0.48 °C and 0.08 °C, manifesting the effectiveness of the proposed method in temperature prediction. Similarly, it can be observed from Fig. 7 (e) that the surface temperature variation range of cell 3 is 34.3 °C to 48.7 °C, and is obviously wider than that of cells 1 and 2. Indeed, its temperature prediction is more difficult than the other two cells. While, from Fig. 7 (d) and (f), it can be observed that the MAE of cell 3 is the same as that of cell 2, and the RMSE and MAX of cell 3 is only slightly larger than those of cell 2, demonstrating the reliability of the proposed method in temperature prediction of aged cell.

In addition, compared to the temperature prediction results for cells 1 to 3 in part A for initial aging, the results for cells 1 and 2 in this part are slightly better than the results for cells 1 and 2 in part A. The reason lies in that the range of temperature variation for cells 1 and 2 in this part is smaller than that in part A, and therefore the prediction accuracy is higher, whereas the case of cell 3 is the opposite of cells 1 and 2, consequently achieving a worse prediction result. From the above analysis, although the surface temperatures of all cells fluctuate and change at different aged periods, the proposed method can track the temperature changes effectively and quickly within an error range of

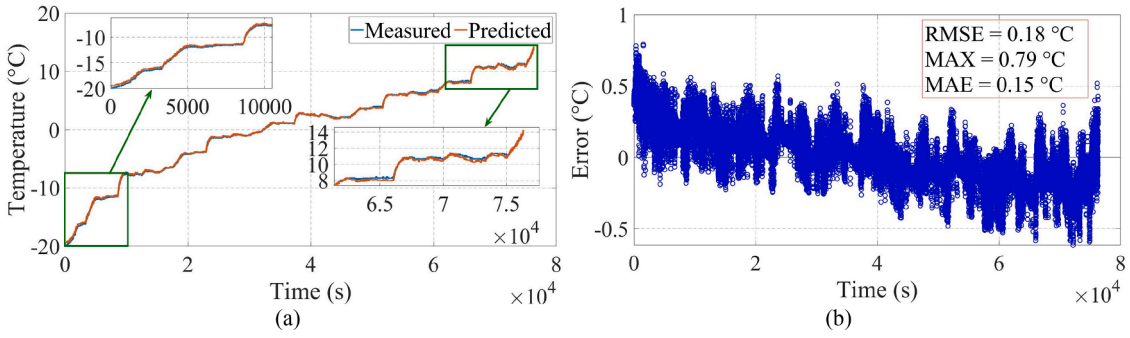


Fig. 6. Temperature prediction results for cell 4. (a) Temperature profiles. (b) Temperature prediction errors.

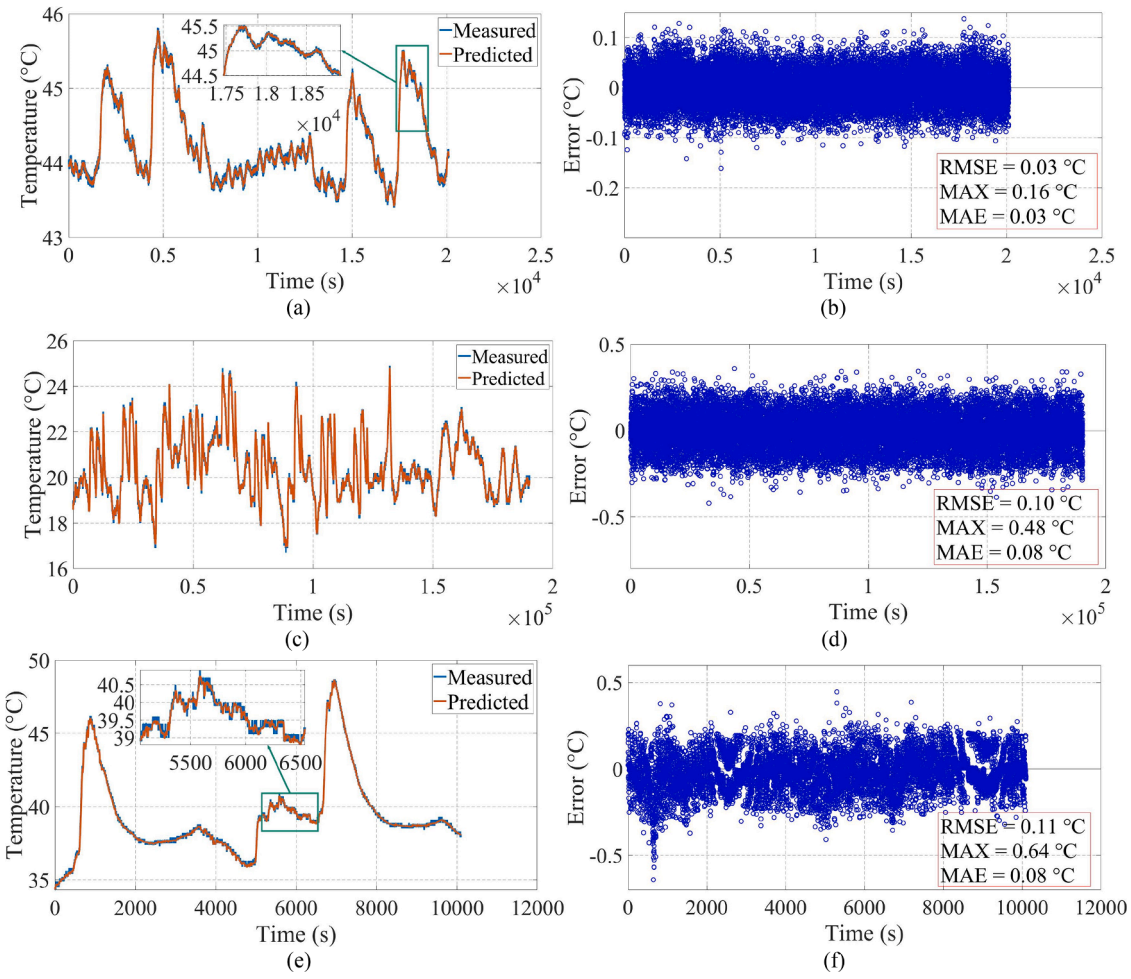


Fig. 7. Temperature prediction results for aged cell. (a) Temperature profiles of cell 1. (b) Temperature prediction errors of cell 1. (c) Temperature profiles of cell 2. (d) Temperature prediction errors of cell 2. (e) Temperature profiles of cell 3. (f) Temperature prediction errors of cell 3.

0.7 °C. The validation results clearly confirm the applicability of the proposed temperature prediction method in a wide life cycle.

4.4. Abnormal temperature diagnosis

During the aging test under driving cycles, cell 3 experienced three temperature anomalies, all of which exceed the maximum allowable working temperature of the battery (60 °C) for long duration. Specifically, the fault data 1 is cut from the charging and discharging data of cell 3 from 864 cycles to 866 cycles, and the abnormal temperature appears in the CV charging stage at 866 cycles and lasts for 215s. Fault data 2 is intercepted from the charge and discharge data of cell 3 at 867 cycles, where the battery abnormal temperature occurs in the CV charging stage at 867 cycles and last for 811s. Fault data 3 is located in

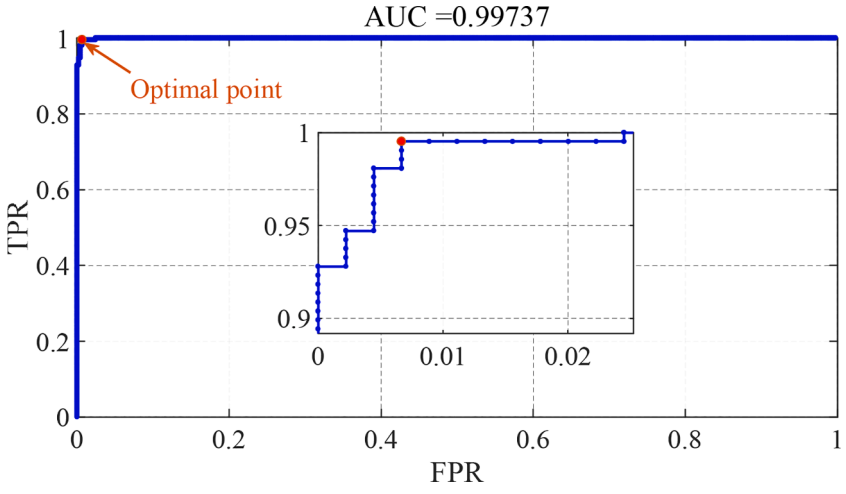


Fig. 8. ROC curve.

the charging and discharging data of cell 3 at 874 cycles, and the temperature exceeds 60 °C during the CV charging stage, and sustains for 671s. These three abnormal temperature data are leveraged to verify the fault diagnosis method proposed in this study. Here, the fault data 1 is hired to determine the optimal threshold, and the ROC curve is illustrated in Fig. 8. It can be found that the value of AUC is 0.997, which is quite close to 1. After calculation, the optimal threshold is -1.68 °C. The corresponding TPR and FPR are 99.5 % and 0.67 %, indicating that when the threshold is set to -1.68 °C, the correct rate of fault diagnosis is as high as 99.5 %, and the false alarm rate is lower than 0.67 %.

The three fault diagnosis results are portrayed in Fig. 9. It can be observed that the temperature of fault data 1 first varies within the normal temperature range from an initial temperature of 43.8 °C, then the temperature reaches 60 °C at 16411s, subsequently the temperature continues to rise to 61.9 °C. When the battery is operated in the normal temperature range, the predicted temperature effectively tracks the changes in the measure. However, when the battery temperature continues to rise from 60 °C to 61.9 °C, the predicted temperature gradually drifts away from the measure. The error distribution in Fig. 9 (b) clearly shows that the residual monitor detects a residual value of -1.75 °C at 16411 s, which exceeds the fault alarm threshold of -1.68 °C, highlighting that the proposed method can effectively diagnose temperature anomalies in a timely manner when it occurs. Moreover, as the temperature continues to rise to 61.9 °C, the absolute residual values detected by the residual monitor increases continuously, demonstrating that the proposed method can sustainably diagnose abnormal battery temperatures.

Analogously, from Fig. 9 (c) and (e), it can be discovered that the temperature variation of fault data 2 and 3 is different from that of fault data 1. The temperature of fault data 2 and 3 first undergoes through a period of normal variation starting at 46.9 °C and 46.8 °C, respectively, then the temperature gradually increases to 66.1 °C and 66.4 °C, and subsequently decreases back to the normal value. Obviously, the range of abnormal temperature variations in fault data 2 and 3 is larger and lasts longer than that of fault data 1. Moreover, it can be seen that the proposed method can accurately track temperature variations within the normal range, while only the temperature variation trend can be predicted on abnormal temperatures. Together with Fig. 9 (d) and (f), one can notice that when the abnormal temperature of fault data 2 and 3 occurs at 3695s and 2968s, their corresponding residual values surpass the fault alarm threshold, indicating that the proposed method can efficiently diagnose the abnormal temperature. Furthermore, when the temperature of fault data 2 varies between 3695 s and 4506 s, the absolute residual value first increases and then decreases as the temperature decreases. When the temperature returns to the normal value at 4506 s, the residual value also returns to the normal value, and the warning is lifted. A similar result also emerges in fault data 3. To sum up, the proposed method cannot only diagnose temperature anomalies reliably, but also relieve fault warning timely after the abnormal temperature returns to the normal value.

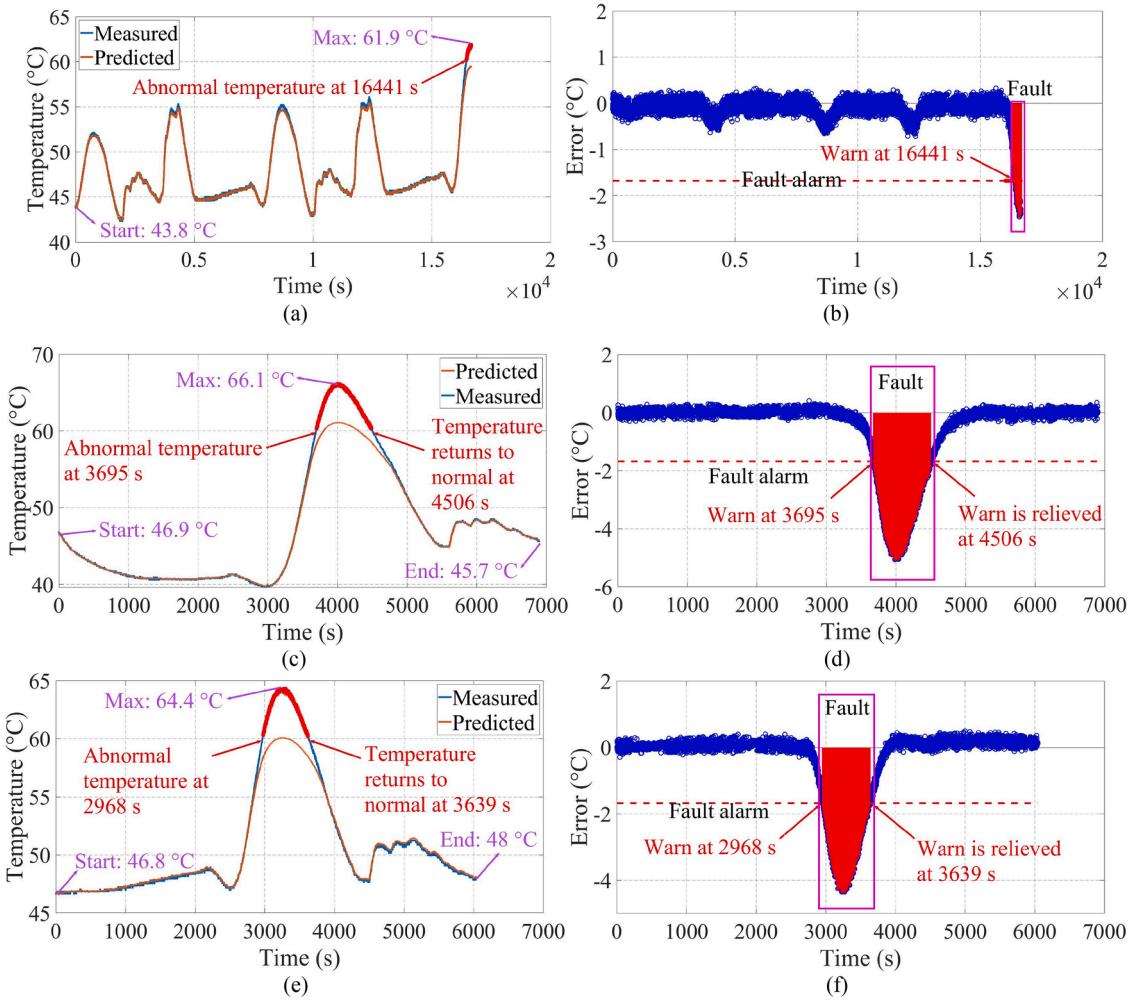


Fig. 9. Abnormal temperature detection results. (a) Temperature prediction profiles of the fault data 1. (b) Residual monitor results of the fault data 1. (c) Temperature prediction profiles of the fault data 2. (d) Residual monitor results of the fault data 2. (e) Temperature prediction profiles of the fault data 3. (f) Residual monitor results of the fault data 3.

In addition, one important concern regarding online applications lies in the computational efficiency [37]. For work that performs online prediction and diagnosis, it is customary to evaluate the computational efficiency based on the time required to run the algorithm. In this case, the total execution time of each prediction and diagnosis costs less than 3.5 ms on a regular computer with the AMD Ryzen 7 3800 × 8-Core Processor 3.89GHz, thus meeting the online application requirement.

5. Conclusions

This study develops a hybrid neural network and fault threshold optimization algorithm to achieve online surface temperature prediction and abnormal temperature diagnosis. First, a hybrid neural network integrating convolutional neural network and long short-term memory (CNN-LSTM) is constructed to excavate the spatial and time-series features in battery operation data and predict the battery temperature online. The proper input parameters of the CNN-LSTM network are determined by evaluating the effect of each parameter of the equivalent circuit thermal model on temperature. Subsequently, the residual value between the online predicted temperature and the measured temperature is calculated in the residual generator, and is further compared with the optimal threshold to determine whether the temperature is abnormal online. Moreover, a fault threshold optimization algorithm based on the receiver operating characteristic curve is employed to select the optimal temperature fault threshold. The accuracy and reliability of the CNN-LSTM network is comprehensively verified using two different types of batteries under dynamic working scenarios including different temperatures and different aged states. By comparing with six commonly machine learning algorithms and four methods in the literature, the CNN-LSTM network shows the highest precision. Furthermore, the verification of three abnormal datasets reveals that the proposed diagnosis strategy can diagnose and unwind temperature fault timely and reliably. Additionally, the total execution time of each prediction and diagnosis is less than 3.5 ms, showcasing the application potential of the proposed method for online battery temperature prediction and abnormal temperature diagnosis.

In the future, the battery internal heat generation mechanism and the internal characteristics of the battery temperature will be investigated in the following work. Meanwhile, a parallel CNN-LSTM network by combining the attention mechanism to achieve a more effective temperature prediction is anticipated in our future research. Additionally, the robustness of the presented method in harsh operation conditions and real-word environment needs to be further addressed.

CRediT authorship contribution statement

Hongqian Zhao: Writing – original draft, Software, Methodology, Conceptualization. **Zheng Chen:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation. **Xing Shu:** Writing – review & editing. **Renxin Xiao:** Writing – review & editing. **Jiangwei Shen:** Writing – review & editing. **Yu Liu:** Writing – review & editing, Supervision. **Yonggang Liu:** Validation, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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